Process & Decision Documentation

Project/Assignment Decisions

For this side quest, I added sounds to the game by using the already implemented sound manager. I decided that for this assignment to not use GenAI and actually go through each file and understand how to make the changes in order to play the sounds. I noticed the sounds were also provided so all I had to do from there was use the event listeners to trigger the sounds when certain actions were taken.

Role-Based Process Evidence

Examples of acceptable process evidence include (but are not limited to):

```
// ----------------------
// SYSTEM listeners (sound/debug)
// ----------------------
if (this.sound) {
  this._unsubs.push(this.events.on("leaf:collected", () => this.sound.play("leaf")));
  this._unsubs.push(this.events.on("player:damaged", () => this.sound.play("hurt")));
  this._unsubs.push(this.events.on("player:died", () => this.sound.play("die")));
  this._unsubs.push(this.events.on("level:won", () => this.sound.play("win")));
  this._unsubs.push(this.events.on("boar:damaged", () => this.sound.play("hit")));
}      You, 7 minutes ago • added files
```

```
//Load sounds
await soundManager.load("hit", "./assets/sfx/hit.wav");
await soundManager.load("leaf", "./assets/sfx/leafcollectwav");
await soundManager.load("win", "./assets/sfx/win.wav");
await soundManager.load("lose", "./assets/sfx/lose.wav");      You, 1
a
```

```
function loadImageAsync(path) {      You, 5 hours ago • added files
  if (!path) {
    // This is the exact scenario that led to GET /undefined.
    throw new Error(`[AssetLoader] loadImageAsync called with invalid path: $
  }
  return new Promise((resolve, reject) => {
    try {
      loadImage(
        path,
        (img) => resolve(img),
        (err) => reject(new Error(`[AssetLoader] Failed to load image "${path
      );
    } catch (e) {
      reject(new Error(`[AssetLoader] loadImage("${path}") threw: ${e?.messag
    }
  });
}
```

```
function loadSoundAsync(path) {
  if (!path) {
    // This is the exact scenario that led to GET /undefined.
    throw new Error(`[AssetLoader] loadImageAsync called with invalid path: $
  }
  return new Promise((resolve, reject) => {
    try {
      loadSound(      You, 1 second ago • Uncommitted changes
        path,   loadSound        function loadSound(path: string | any[], suc…
        (img) = loadSoundAsync
        (err) => reject(new Error(`[AssetLoader] Failed to load image "${path
      );
    } catch (e) {
      reject(new Error(`[AssetLoader] loadImage("${path}") threw: ${e?.messag
    }
  });
}
```

```
// - This keeps audio concerns isolated from gameplay and suppor
import {loadSoundAsync} from "./AssetLoader.js";
You, 8 seconds ago | 1 author (You)
export class SoundManager {
  constructor() {
    this.sfx = {};
  }

  async load(name, path) {
    this.sfx[name] = await loadSound(path);
  }

  play(name) {
    this.sfx[name]?.play();
  }      You, 5 hours ago • added files
}
```

```
top ▾   Filter                                   Default levels ▾   No Issues
BOOT: start                                                          main.js:121
▶GET http://127.0.0.1:5500/assets/sfx/hit.wav 404 (Not Found)        p5.sound.min.js:2
▶BOOT FAILED: Error: [AssetLoader] Failed to load sound "./assets/sfx/hit.wav": loadSound:  main.js:240
Unable to load ./assets/sfx/hit.wav. The request status was: 404 (Not Found)
    at AssetLoader.js:184:25
    at XMLHttpRequest.<anonymous> (p5.sound.min.js:2:103030)
> cmd i to turn on code suggestions. Don't show again
```

```
await soundManager.load("hit", "./assets/sfx/hitEnemy.wav");      You, 1 sec
```

```
// ---------------------
if (this.sound) {
  this._unsubs.push(this.events.on("leaf:collected", () => this.sound.play("leaf")));      You, 5
  this._unsubs.push(this.events.on("player:damaged", () => this.sound.play("hurt")));
  this._unsubs.push(this.events.on("player:jumped", () => this.sound.play("jump")));
  this._unsubs.push(this.events.on("level:won", () => this.sound.play("win")));
  this._unsubs.push(this.events.on("boar:damaged", () => this.sound.play("hit")));
}
```

```
export function loadSoundAsync(path) {      You, 1 second ago • Uncommitted changes
  if (!path) {
    // This is the exact scenario that led to GET /undefined.
    throw new Error(`[AssetLoader] loadSoundAsync called with invalid path: ${path}`);
  }
  return new Promise((resolve, reject) => {
    try {
      loadSound(
        path,
        (sound) => resolve(sound),
        (err) => reject(new Error(`[AssetLoader] Failed to load sound "${path}": ${err}`)),
      );
    } catch (e) {
      reject(new Error(`[AssetLoader] loadSound("${path}") threw: ${e?.message ?? e}`));
    }
  });
}
```

```
// for the background music      You, 1 second ago • Uncommitted changes
this.sound.play("music");
```

```
//Load sounds
await soundManager.load("hit", "./assets/sfx/hit.wav");
await soundManager.load("leaf", "./assets/sfx/leafcollect.wav");
await soundManager.load("jump", "./assets/sfx/jump.wav");
await soundManager.load("music", "./assets/sfx/music.wav");
await soundManager.load("hurt", "./assets/sfx/receieve.wav");
a
```

```
//Load sounds
await soundManager.load("hit", "./assets/sounds/hit.wav");
await soundManager.load("jump", "./assets/sounds/jump.wav");
```

```
function loadImageAsync(path) {
  if (!path) {
    // This is the exact scenario that led to GET /undefined.
    throw new Error(`[AssetLoader] loadImageAsync called with invalid path: ${path}`);
  }
  return new Promise((resolve, reject) => {
    try {
      loadImage(
        path,
        (img) => resolve(img),
        (err) => reject(new Error(`[AssetLoader] Failed to load image "${path}": ${err}`)),
      );
    } catch (e) {
      reject(new Error(`[AssetLoader] loadImage("${path}") threw: ${e?.message ?? e}`));
    }
  });
}
```

Goal of Work Session

The goal of this work session was to add some type of audio feedback into the game to enhance the player experience. Since sounds were all provided, I had to only change and alter the code in order to achieve this.

GenAI Documentation

No GenAI was used for this assignment.

Summary of Process (Human + Tool)

Since I chose not to use AI, this had taken me quite a while as I was writing my code by hand. First, before doing anything, I made sure to read carefully through all of the files to understand what was happening. I noticed certain parts of the code that would be helpful that were being used for loading other assets, such as images, and figured that they could help me when adding in the sounds. Since I was writing the code by hand throughout the process, I noticed a few mistakes that would result in breaking the game, but in order to debug, I would inspect in the Chrome browser and play test on a live server. Most of the time, there were errors from misspelling a word or forgetting to write the complete name. At times, I relied on Co-pilot's automatic suggestions, but they would usually just give something random and hallucinate, which would also cause it to

break. This resulted in me doing it myself in the end, just to be sure that everything was working properly and referring to the correct files when loading anything.

Verification & Judgement

I kept playtesting myself on a live server to double-check if my changes were resulting in any differences to the game. This way, I would keep going through the process of iteration until I was able to get the sound working. At any time, if I added something wrong or messed it up, I used the inspect in the Chrome browser to see clearly what the error was. Usually, if there were any errors, the game would break, as most of the struggles I was having were loading the sounds, since most of my mistakes did not refer to the correct names for the files.

Limitations, Dead Ends, or Open Questions

Since I only implemented sounds, I was wondering what other visual feedback or changes I could add that have not already been included in the game. Since the game is fairly straightforward, this was considered not to add any new mechanics but rather for triggers.